MILITARY SPECIFICATION

MOLDING COMPOUND, HOUSING

This specification is approved for use within the US Army Armament, Munitions and Chemical Command, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one type of two component silica-filled epoxy molding compound designated as Molding Compound, Housing.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise-specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto; cited in the solicitation.

SPECIFICATIONS

MILITARY

MIL-A-48078 - Ammunition, Standard Quality Assurance Provisions, General Specification For.

STANDARDS

FEDERAL

FED-STD-595 - Colors.

MILITARY

MIL-STD-105 - Sampling Procedures and Table for Inspection by Attributes.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to Commander, U S Army Armament, Munitions and Chemical Command, Attn AMSMC-QA Picatinny Arsenal, New Jersey 07806-5000 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter

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2. 1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

DRAWINGS (See 6.8)

US ARMY ARMAMENT RESEARCH, DEVELOPMENT AND ENGINEERING CENTER (ARDEC)

9277699	- Silane
9277700	- Chlorendic Anhydride
9377701	- Anhydride
9277704	- Maleic Anhydride
9298447	- Silica, Granular
9298444	- Silica, Powdered

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.1.3 Other publications. The following document(s) form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted shall be those listed in the issue of the DODISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS shall be the issue of the nongovernment documents which is current on the date of the solicitation.

AMERICAN SOCIETY FOR TESTING AND MATERIALS

ASTM-B-214	- Method of Test for Sieve Analysis of Granular Metal Powders.
ASTM-D-257	- D-C Resistance or Conductance of Insulating Materials
ASTM-D-618	- Conditioning Plastics and Electrical Insulating Materials for Testing.
ASTM-D-648	- Deflection Temperature of Plastics Under Flexural Load.
ASTM-D-695	- Compressive Properties of Rigid Plastics.
ASTM-D-790	- Flexural Properties of Plastics.
ASTM-D-792	- Specific Gravity and Density of Plastics by Displacement.
ASTM-D-2471	- Gel Time and Peak Exothermic Temperature of Reacting Thermosetting Resins.

(Application for copies of ASTM publications should be addressed the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

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2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets of MS standards) , the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Material. The material shall consist of two components, the base resin and the hardener. Each component and the mixture shall conform to its respective requirements of this specification. The base resin and the hardener shall be made only of the materials listed in 3.1.1 and 3.1.2 respectively.

3.1.1 Base resin composition (Part A) (See 6.6).

Ingredient

Epoxy Novolac Resin
Epoxide equivalent weight 176-181
(Dow Chemical Co. Den 438 or equivalent)

2, 3 - Epoxy propanol (Glycidol)

Gamma-Glycidoxypropyltrimethoxy Silane (Dwg. 9277699)

Silica, Granular (Dwg. 9298447)

Silica, Powdered (Dwg. 9288444)

Foam Reducer
(Dow Corning Antifoam A or equivalent]

3.1.2 Hardener composition (Part B) (See 6.6).

Ingredient

Maleic Anhydride (Dwg. 9277704)

Chlorendic Anhydride (Dwg. 9277700)

Methyl Tetrahydrophthalic Anhydride (Dwg. 9277701)
(Union Carbide ZZL0334 or equivalent)

Dipropylene Glycol

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Uranyl Acetylacetonate (99% Purity by chemical analysis-advisory)
(Orion Chemical Company, Provo, Utah)

Gamma-Glycidoxypropyltrimethoxy Silane (Dwg. 9277699)

Silica, Granular (Dwg. 9298447)

Silica, Powdered (Dwg. 9298444)

Foam Reducer
(Dow Corning Antifoam A or equivalent)

3.2 Base resin viscosity. The viscosity of the base resin when measured at 70°F to 75°F in accordance with 4.5.1, a minimum of 72 hours after the material is first compounded, shall not exceed 2.6×10^6 centipoises.

3.3 Hardener viscosity. The viscosity of the hardener when measured at 70°F to 75°F in accordance with 4.5.2, a minimum of 72 hours after the material is first compounded, shall not exceed 6.5×10^5 centipoises.

3.4 Molding compound. The molding compound, when prepared as specified in 4.5.3.1 and 4.5.4.1, shall comply with the requirements specified in Table I and Table II when tested as specified in the referenced test paragraphs.

The requirements apply to molding compound made from base resin (Part A) and hardener (Part B) that have each been "wet out" (aged) for seventy-two (72) hours minimum from the time they are first compounded and prior to mixing together in the specified ratios to make the molding compound.

TABLE I

Mixed molding compound - within 5 minutes after mixing

Test	Requirement	Test Paragraph Reference
Viscosity, cps, max	800,000	4.5.3.2
Gel time, minutes, max	20 at 250°F	4.5.3.3
Peak exotherm, max	10°F at 160°F	4.5.3.4

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TABLE II

Cured molding compound

Test	Requirement	Test Paragraph Reference
Specific Gravity	2.1 minimum	4.5.4.2
Compressive Strength, psi	32,000 minimum	4.5.4.3
Compressive Modulus of elasticity, psi	2.3 - 3.8 x 10^6	4.5.4.4
Flexural Strength, psi	17,000 minimum	4.5.4.5
Flexural Modulus of Elasticity, psi	2.80 to 3.60 x 10^6	4.5.4.6
Area under flexure stress/strain curve	26.5 in. lb/sq. in. min	4.5.4.7
Insulation resistance, ohms	1 x 10^{11} minimum	4.5.4.8
Coefficient of Linear Thermal Expansion		4.5.4.9
75°F to 165°F (23.9C to 73.9°C)	40 x 10^{-6} in/in/°C, max	
-65°F to 75°F (-53.9C to 23.9°C)	25 x 10^{-6} in/in/°C, max	
Heat Distortion Temperature	158°F minimum	4.5.4.10

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3.5 Color. The color of cured molding compound shall be within the range of colors bounded by color numbers 34079 and 34102 inclusive of FED-STD-595.

3.6 Data requirements. The contractor shall generate data in accordance with the requirements of the data item descriptions cited in 4.3.2.1, 4.4.2.1, 4.4.2.2, 4.4.2.3, 4.4.2.4 and 4.4.2.5.

3.7 Labeling of containers Each container used to ship the base resin and hardener shall be suitably labeled with warnings that the contents contain toxic substances. (See 6.4).

3.8 First article testing. This specification contains technical provisions for first article testing. Requirement for submission of first article samples by the contractor shall be as specified in the contract.

3.9 Workmanship.

3.9.1 Silica. The silica shall be uniform in quality and free from dirt, metallic particles and other foreign material.

3.9.2 Base resin and hardener. The base resin and hardener shall be free of lumps and foreign material. The containers shall be wide-mouthed, properly labeled, free of damage, and there shall be no evidence of leakage or improper filling.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements. Unless otherwise specified herein or in the contract, the provisions of MIL-A-48078 shall apply and are hereby made a part of this detail specification.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

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4.2 Classification of inspections. The inspection requirements specified herein are classsified as follows:

a. First article inspection (See 4.3).

b. Quality conformance inspection (See 4.4).

4.3 First article inspection.

4.3.1 Submission. The contractor shall submit a first article sample as designated by the Contracting Officer for evaluation in accordance with provisions of 4.3.2. The first article sample shall consist of the following items in sample quantities as indicated below and Table I.

a. one (1) container of the base resin.

b. One (1) container of the hardener.

4.3.2 Inspection to be performed. See MIL-A-48078 and Table I and II as specified herein.

4.3.2.1 Base resin and hardener components.

a. Certifications shall be provided for the examinations and tests specified in 4.4.2.1 and 4.4.2.2 for the two silicas, and shall include complete test results for the particle size distribution.

b. Certification shall be provided for formulation completeness and shall include the actual. weights or percentages by weight of all constituents used in formulating the base resin.

c. One sample of the base resin shall be tested for viscosity as specified in 4.5.1.

d. Certification shall be provided for formulation completeness and shall include the actual weights or percentages by weight of all constituents used in formulating the hardener.

e. One sample of the hardener shall be tested for viscosity as specified in 4.5.2.

4.3.2.2 Molding compound. After completion of the tests for base resin and hardener viscosity a sufficient quantity of molding compound shall be prepared in accordance with 4.5.3.1 and subjected to the tests listed in Table I. Testing shall be as specified in 4.5.3.2 through 4.5.3.4. Test specimens will also be prepared in accordance with 4.5.4.1 and subjected to the tests listed in Table II. Testing shall be as specified in 4.5.4.2 through 4.5.4.10.

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4.3.3 Rejection. See MIL-A-48078. In addition, the first article quantity shall be rejected if any of the following occur:

a. The certification for either of the two silicas reveal any failure to comply with any of the requirements specified on the applicable drawing.

b. The certifications for the base resin or the hardener is incomplete or indicates any constituent is missing.

c. The base resin or hardener exceeds the maximum viscosity.

d. The "molding compound tested fails to comply with any of the requirements listed in Table I and Table II, or fails to comply with the color requirement of 3.6.

.4 Quality conformance inspection.

4.4.1 Inspection lot formation. Inspection lots shall comply with the lot formation provisions of MIL-A-48078. In addition, inspection lots of base resin or hardener shall consist of that quantity of material produced from the same batch of raw material and which is both manufactured and offered for acceptance at one time.

4.4.2 Examination. See MIL-A-48078.

a. Sampling plans. Unless otherwise specified in the Classification of Defects and Testing, sampling plans for major and minor defects shall be in accordance with MIL-STD-105 Inspection Level II.

b. Inspection. Material samples in accordance with 4.4.2.a for Quality Conformance Inspection shall conform to the requirements of 4.4.2.1 to 4.4.2.6.

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.1	Silica, Granular				9298447
					NEXT HIGHER ASSEMBLY --
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Particle size distribution			3.1	4.5.5.1 (Certification)
102	Weight loss on ignition			3.1	4.5.5.2 (Certification)
103	Silicon dioxide content			3.1	4.5.5.3 (Certification)
Minor					
201	Presence of any foreign material	Each container		3.9.1	Visual
	If any sample fails to comply with any of the applicable requirements, the silica represented by the sample shall be rejected for use in the resin or hardener.				

NOTES: 1. Certifications shall include complete test results (variables data) for the particle size distribution. Certifications shall be submitted in accordance with Data item Description DI-R-1724 for each lot.

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CLASSIFICATION OF DEFECTS & TESTS

MIL-M-48245B(AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9298444
4.4.2.2	Silica, Powdered				NEXT HIGHER ASSEMBLY --
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Particle size distribution			3.1	4.5.5.1 (Certification)
102	Weight loss on ignition			3.1	4.5.5.2 (Certification)
103	Silicon dioxide content			3.1	4.5.5.3 (Certification)
Minor					
201	Presence of any foreign material	Each container		3.9.1	Visual
	If any sample fails to comply with any of the applicable requirements, the silica represented by the sample shall be rejected for use in the resin or hardener.				

NOTES: 1. Certifications shall include complete test results (variables data) for the particle size distribution. Certifications shall be submitted in accordance with Data item Description DI-R-1724 for each lot.

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CLASSIFICATION OF DEFECTS & TESTS

MIL-M-48245B(AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.3	Base Resin (Part A)				--
					NEXT HIGHER ASSEMBLY
					--
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Viscosity	Note 1		3.2	4.5.1
102	Completeness of formulation	Note 2		3.1.1.	Certification
Minor					
201	Poor workmanship	Note 1		3.9.2	Visual

NOTES: 1. Sample size shall be 1 container per lot (Reference paragraph 4.4.1). No defect permitted.
2. Reference paragraph 4.3.2.1.b.

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CLASSIFICATION OF DEFECTS & TESTS

MIL-M-48245B(AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER ---
4.4.2.4	Hardener (Part B)				NEXT HIGHER ASSEMBLY --
CATEGORY	EXAMINATION OR TEST	NO OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Viscosity	Note 1		3.3	4.5.2
102	Completeness of formulation	Note 2		3.1.2	Certification
Minor					
201	Poor workmanship	Note 1		3.9.2	Visual

NOTES: 1. Sample size shall be 1 container per lot (Reference paragraph 4.4.1). No defect permitted.

2. Reference paragraph 4.3.2.1.d.

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CLASSIFICATION OF DEFECTS & TESTS

MIL-M-48245B(AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.5	Molding Compound	1 OF 1	--
			NEXT HIGHER ASSEMBLY: --

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major		Notes			
101	Viscosity	1		3.4	4.5.3.2
102	Specific gravity	1		3.4	4.5.4.2
103	Compressive strength	2		3.4	4.5.4.3 Note 4
104	Compressive modulus	2		3.4	4.5.4.4
105	Flexural strength	2		3.4	4.5.4.5
106	Flexural modulus of elasticity	2		3.4	4.5.4.6
107	Area under flexure stress/strain curve	2		3.4	4.5.4.7
108	Color correct	2		3.6	Visual
Minor					
201	Poor workmanship (presence of lumps or foreign matter)	3		3.9.2	Visual

NOTES:

1. The average value of three (3) tests shall be used.
2. The average value of five (5) specimens shall be used.
3. Examine material during mixing and while casting test specimens.
4. Variables Data for all tests shall be submitted in accordance with Data Item Description DI-R-01724.

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-M-48245B(AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.6	Container, sealed				--
					NEXT HIGHER ASSEMBLY
					--
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%,	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical					
1	Warning label missing (See 5.2, 6.4, and 6.5)		100%	3.8	Visual
Major					
101	Container not wide-mouthed		0.40%	3.9.2	Visual
Minor					
201	Container not sealed		0.65%	3.9.2	Visual
202	Leakage		0.65%	3.9.2	Visual
203	Container label incorrect or incomplete		0.65%	3.9.2	Visual

NOTES:

AMSMC(m 1570, 1 Feb 85 Replac(RSMC-QA (D) Form 160, 1 Aug 83, which may not be u

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4.4.3 Testing. Testing is described in First Article testing and Quality Conformance Inspection Tables.

4.4.4 Inspection equipment. The inspection equipment required to perform the examinations and tests prescribed herein is described in the "Paragraph Reference/Inspection Method" column in the tables starting with 4.4.2.1. The contractor shall submit for approval inspection equipment designs in accordance with the terms of the contract. See Section 6 of MIL-A-48078.

4.5 Methods of inspection. (See 6.5)

4.5.1 Base resin viscosity. Base resin viscosity shall be determined using a Brookfield RVF Helipath viscosimeter with a T - f spindle at a speed of 4 rpm. The viscosity reading shall be made at the tenth revolution of the spindle.

4.5.2 Hardener viscosity. Hardener viscosity shall be determined using a Brookfield RVF Helipath viscosimeter with a T - F spindle at a speed of 10 rpm. The viscosity reading shall be made at the tenth revolution of the spindle.

4.5.3 Molding compound properties - immediately after mixing.

4.5.3.1 Compound mixing. Immediately prior to mixing, the base resin container shall be agitated on a paint shaker for 45 minutes minimum and the hardener container shall be agitated on the shaker for 30 minutes minimum. The mixed molding compound for use in determining conformance to Table I shall be prepared by thoroughly mixing two (2) parts ± 5% by weight of the base resin with one (1) part ± 5% by weight of hardener for not more than five (5) minutes, and immediately degassing at an absolute pressure of 2 to 4 torr for 3 ± 1 minutes at a temperature of 150° ± 10°F. The mix shall be used in testing for the requirements of Table I within 5 minutes after mixing.

4.5.3.2 Viscosity of molding Compound. The viscosity of each lot of molding compound shall be determined on material mixed and degassed as specified in 4.5.3.1 except that the material shall be mixed at room temperature. Viscosity-shall be determined by using a Brookfield RVF Helipath viscosimeter with a T-F spindle at a speed at 10 rpm, at a temperature of 75° to 80°F. The viscosity shall be recorded after 10 revolutions of the spindle.

Three (3) viscosity measurements shall be made on each batch. If the average value of the three viscosity readings fails to comply with the value specified in Table I the molding compound shall be rejected.

4.5.3.3 Gel time. Gel time shall be determined on a batch of compound mixed and degassed as specified in 4.5.3.1. Gel time shall be determined using the methods and procedures specified in ASTM-D-2471.

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4.5.3.4 Peak exotherm. The peak exotherm produced by mixing the components as specified in 4.5.3.1 shall be measured with a thermocouple centered in a 100 cubic centimeter volume contained in a three ounce cup, placed in a bath at 160°F and using the methods and procedures specified in ASTM-D-2471.

4.5.4 Molding compound properties after curing.

4.5.4.1 Specimen preparation. Test specimens shall be cast in suitable molds, as required for the testing for conformance to Table II. Resin and hardener shall be prepared,-mixed and degassed as specified in 4.5.3.1, then poured into the molds. Test bars shall be cast using best commercial practice including vibration and evacuation as necessary. The material shall be cured for 40 ± 1/2 hours at 160° ± 5°F.

4.5.4.2 Specific gravity. The specific gravity shall be determined in accordance with ASTM-D-792, Method A-1, on three (3) samples approximately 1.5 inches long taken from test bars cast without overflow in vertical 1/2 inch by 1/2 inch by 6 inch molds and allowed to cool slowly.

4.5.4.3 Compressive strength. The compressive strength shall be determined as specified in ASTM-D-695. The specimens shall be a right prism with dimensions 0.5 inch by 0.5 inch-by 1.6 inches. Five specimens shall be tested. The end faces of each specimen shall be flat, parallel to each other and perpendicular to the sides to within .003 inches total variation. The cross sectional area of each specimen shall be determined by measuring the two 0.5 dimensions within .001. All dimensional measurements shall be made immediately prior to testing. Specimens shall be conditioned and tested as specified in the above test method. A compressive force/compressive deformation chart shall be made for each specimen using a deflectometer and suitable automatic recording device. The sensor of the deflectometer shall be placed adjacent to the specimen so as to exclude machine related deflections from the deflection measurement. In addition, a deflection correction factor shall be obtained for the deflection measurement by determining the inherent deflection in the equipment when the full-scale load is applied to a specimen of the same length as the test specimen, but which has negligible deflection under the full scale load. (This correction factor is necessary for determining the compressive modulus in 4.5.4.4.) The compressive strength for each test specimen shall be calculated by dividing the maximum compressive load carried by the specimen by the original cross sectional area of the specimen. If the average compressive strength of the five specimens fails to comply with the applicable requirement in Table II the molding compound shall be rejected.

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4.5.4.4 Compressive modulus of elasticity. The compressive modulus of elasticity shall be determined using the chart from the compressive force test for each of the five compression specimens. A tangent line shall be drawn on the initial linear portion of the curve. A point along this line and common to the curve shall be selected and the corresponding total deformation at this point shall be determined. This value shall be reduced by using the correction factor (in/1,000 lb) obtained in 4.5.4.3 to obtain the true total deformation of the specimen. The unit strain (in\in) shall be determined by dividing the true specimen deformation (inches) by the original specimen length (inches). The compressive modulus shall then be calculated by dividing the load (pounds) at the selected point on the curve by the unit strain (in/in) and dividing the result by the original cross sectional area of the specimen (square inches). If the average compressive modulus of the five specimens fails to comply with the applicable requirements in Table II, the molding compound shall be rejected.

4.5.4.5 Flexural strength. The flexural strength shall be determined as specified in ASTM-D-790, Method 1. Specimen sizes shall be 1/2 inch by 1/2 inch x 5 inches and a 4-inch span shall be used. Five specimens shall be tested. The flexural strength S, shall be calculated as follows:

$$S = 3PL/2bd^2 \quad \text{(psi)}$$

where

P= maximum load (pounds)

L = support span (inches)
b = width of the specimen (inches)
d = depth of the specimen (inches)

The width and depth of the specimen shall be measured to within .001 inches. If the average flexural strength of the five specimens fails to comply with the applicable requirement in Table II the molding compound shall be rejected.

4.5.4.6 Flexural modulus of elasticity. The flexural modulus of elasticity shall be determined as specified in ASTM-D-790, Method 1, from the load deflection chart generated in the flexural strength tests. Flexural modulus, E, shall be calculated as follows:

$$E = L^3m/4bd^3 \quad \text{(psi)}$$

where

m = slope of the tangent to the initial straight line portion of the curve (lb/in)

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L, b, and d are as stated for the flexural strength test. If the average flexural modulus of elasticity for the five specimens fails to comply with the applicable requirement in Table II, the molding compound shall be rejected.

4.5.4.7 Area under the flexural load/deformation curve. The area under the flexural load/deformation curve shall be determined using a polar planimeter (Gleman Instrument Co., or equivalent) and the curve from the flexural strength test in 4.5.4.5. The area under the curve shall be determined by tracing the entire 3 sides of the "triangular shaped" figure formed by the curve on the load/deformation chart. A constructed line shall be drawn perpendicular to the deformation axis of the chart to intersect the failure point on the curve to form the triangular figure. The area of the figure in square inches, must then be multiplied by scale factors on the chart (lb/in) and (in/in) to obtain the test value in inch pounds (in-lb) .

This value shall then be divided by the cross-sectional area of the test specimen as determined by multiplying the width and depth measurements obtained in 4.5.4.5 for the specimen. The result is the value for the area under the flexure stress/strain curve. If the average of the five values obtained by this procedure for the five specimens fails to comply with the applicable requirement in Table 11 the molding compound shall be rejected.

4.5.4.8 Insulation resistance. The DC insulation resistance of three specimens shall be determined in accordance with ASTM-D-257. The specimens shall conform to C of figure 3 in the above test method, except that specimens shall be 1/2" by 1/2". The specimens shall be conditioned in accordance with Procedure C of ASTM-D-618 prior to measuring the resistance. The test shall be performed using a commercial megohm meter.

4.5.4.9 Coefficient of linear thermal expansion. The coefficient of linear thermal expansion shall be determined by thermal mechanical analysis, or approved equivalent. Two specimens shall be tested at each temperature range specified in Table II. (Total of four specimens.) Specimen size shall be either 1/8 inch by 1/8 inch by 0.2 to 0.3 inch thick, or 1/2 inch diameter by 0.2 to 0.3 inch thick. The heating rate shall be 10°C per minute. Prior to testing, the specimens shall be conditioned as specified in procedure A of ASTM-D-618.

4.5.4.10 Heat distortion temperature. The heat distortion temperature shall be determined on three specimens conditioned as specified in Procedure A of ASTM-D-618. Testing shall be in accordance with ASTM-D-648. An acceptable alternative is thermal mechanical analysis using test specimens approximately 0.125 inch by 0.125 by 0.50 inch thick. If any specimen fails to comply with,the applicable requirement in Table II, the molding compound shall be rejected.

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4.5.5 Silica properties.

4.5.5.1 Particle size distribution. The silica particle size distribution shall be determined in accordance with ASTM-B-214 using the required sieves.

4.5.5.2 Loss on ignition. The loss on ignition shall be determined by drying a sample of the lot of silica at 200° ± 10°F for 24 hours minumum. A 5 gm sample shall then be accurately weighed placed in a muffle furnace and ignited at 950° ± 15°C for a minimum of two (2) hours. The sample shall be cooled in a desiccator and reweighed. The percent of loss on ignition shall be calculated as follows:

$$\frac{\text{Original weight - Final weight}}{\text{Original weight}} \times 100$$

4.5.5.3 Silicon dioxide content. The silicon dioxide content may be determined by the wet analysis method as follows:

Weigh accurately about 0.5 gram of the material. Transfer to a 250 ml beaker and boil for 30 minutes with 50 ml of hydrochloric acid (1:1). Add 50 ml water. Filter on a double ashless filter paper . Wash. Transfer filter and residue to a tared platinum dish. Ignite. Weigh the insoluble residue. The insoluble residue shall then be volatilized with 0.5 ml of sulfuric acid and 10 ml (1:1) hydrofluoric acid. (Add the hydrofluoric acid very cautiously to avoid mechanical loss from the vigorous reaction set up between HF and SiO_2.) Evaporate under a hood. Ignite to drive off white fumes, then weigh. The loss in weight in the insoluble residue from volatilization will be the total silica (SiO_2). Repeat acid digestion until weight loss on repetitive trials is less than 2% difference.

The contractor may use x-ray analysis or other suitable test method if approved by the Contracting Officer and submitted as specified in 4.4.4.

5. PACKAGING

5.1 Packaging. The base resin and hardener shall each be packaged in widemouthed containers as specified by the procuring agency.

5.2 Marking. All marking shall be clear and legible and shall include as a minimum:

Manufacturer’s name
Item identification and lot number
Date of manufacture
Title and date of revision of this Military Specification
Safety precaution warnings

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5.3 Shipping. When molding compound from more than one lot is shipped at one time, each lot shall be kept separate, and the division between lots clearly indicated to prevent mixing of the lots in transit.

6. NOTES

6.1 Intended use. The material covered by this specification is intended for use in molding munitions and plugs for the Projectile, 155MM, M692/M731.

6.2 Ordering data. See MIL-A-48078.

6.3 Submission of inspection equipment designs for approval. (See 6.2.3 of MIL-A-48078) Submit equipment designs as required to Commander, AMCCOM, ATTN: AMSMC-QAT-I(D), Picatinny Arsenal, NJ 07806-5000. This address will be specified on the Contract Data Requirements List, DD Form 1423 in the contract. Unless otherwise specified, data item DI-R-1714 will apply.

6.4 Safety precautions. Since some of the constituents and compounds are toxic and may exude toxic fumes when mixed or heated, suitable precautions must be taken when handling, mixing, or curing this molding compound or its constituents. Use adequate ventilation. Avoid contact with skin, eyes, and clothing. Avoid prolonged or repeated breathing of fumes.

6.5 Shelf life. The shelf life of both the base resin (part A) and the hardener (Part B) is nine (9) months from the date of manufacture. Material which has passed the shelf life expiration date may be tested for extension of shelf life. Tests shall consist of 4.3.2.1.c, 4.3.2.1.e, 4.4.2.5 and 4.4.2.6 If the material meets the requirements of these tests as specified in Tables I and II, shelf life may be extended for three (3) months from the date Of test. After the expiration of the first shelf life extension, materials may be tested for additional shelf life extensions. Tests shall consist of 4.3.2.1.c, 4.3.2.1.e, 4.4.2.5 and 4.4.2.6. If the material meets the requirements of these tests as specified in Tables I and II, shelf life may be extended for one (1) month from the date of retest.

6.6 Composition. The following amounts of each constituent (percentage of weight) have been used to make the resin and hardener. These nominal amounts may be adjusted, provided that all the requirements of section 3 are met.

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Base resin composition (Part A)

Ingredient	Percent by weight
Epoxy Novolac Resin Epoxide equivalent weight 176-181 (Dow Chemical Co. Den 438 or equivalent)	17.39
2, 3 - Epoxy propanol (Glycidol)	3.07
Gamma-Glycidoxypropyltrimethoxy Silane (Dwg. 9277699)	1.37
Silica, Granular (Dwg. 9298447)	53.20
Silica, Powdered (Dwg. 9288444)	24.96
Foam Reducer (Dow Corning Antifoam A or equivalent)	0.01

Hardener composition (Part B)

Ingredient	Percent by weight
Maleic Anhydride (Dwg. 9277704)	4.80
Chlorendic Anhydride (Dwg. 9277700)	6.41
Methyl Tetrahydrophthalic Anhydride (Dwg. 9277701)	9.58
Dipropylene Glycol	0.41
Uranyl Acetylacetonate (99% Purity)	0.30
Gamma-Glycidoxypropyltrimethoxy Silane (Dwg. 9277699)	1.35
Silica, Granular (Dwg. 9298447)	52.37
Silica, Powdered (Dwg. 9298444)	24.77
Foam Reducer (Dow Corning Antifoam A or equivalent)	0.01

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6.7 Data requirements. Deliverable data required by this specification is cited in the following paragraphs:

Paragraph	Date Requirement	Applicable Did
4.3.2.1	Certifications	DI-R-1724 (Tailored)
4.2.2.1	Certifications	DI-R-1724 (Tailored)
4.4.2.2	Certifications	DI-R-1724 (Tailored)
4.4.2.3	Certifications	DI-R-1724 (Tailored)
4.4.2.4	Certifications	DI-R-1724 (Tailored)
4.4.2.5	Test results	DI-R-1724
6.9	Ammunition Data Card	DI-L-1410

(Copies of the data item descriptions required by the contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the Contracting Officer.)

6.8 Drawings. Drawings listed in Section 2 of this specification under the heading US Army Armament Research, Development and Engineering Center (ARDEC) may also include drawings prepared by, and identified as US Army Armament Research and Development Command, Edgewood Arsenal, Frankford Arsenal, Rock Island Arsenal, or Picatinny Arsenal drawings. Technical data originally prepared by these activities is now under the cognizance of ARDEC.

6.9 Submission of test data. In addition to the normal distribution of records, when the molding compound is procured by the US Army Armament Munitions and Chemical Command (AMCCOM), one (1) copy of all Ammunition Data Cards shall be forwarded to Commander, ARDEC, ATTN: AMSMC-QAT-M(D) , Dover, NJ 07801-5001.

6.10 Subject term (key word) listing.

Molding compound
Housing
Base resin
Hardener
Silica
Specific gravity
Compression strength
Compression modules of elasticity
Flexural strength
Flexural modules of elasticity
Insulation resistance
Coefficient of linear thermal expansion
Heat distortion temperature
Shelf life

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6.11 Changes from previous issue. Asterisks (or vertical lines) are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodian:
Army-AR

Preparing activity:
Army-AR

(Project 1345-A525)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-M-48245B (AR)	MOLDING COMPOUND, HOUSING

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*: _____
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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DD FORM 82 MAR 1426 PREVIOUS EDITION IS OBSOLETE.

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